



Final Software Requirements Specification for CarbonTrack: SOAST Students Carbon Footprint Monitoring System

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1. Project Charter

This section establishes the overarching foundation and direction of the CarbonTrack project. It clarifies the primary reasons for developing the system, the criteria used to measure its success, and the potential risks the team must prepare for.

1.1 Objectives

The primary objectives of the CarbonTrack system aim to directly resolve the operational inefficiencies, validation bottlenecks, and reporting limitations of the institution's existing manual environmental monitoring process:

Digitization & Centralization: To transition the SOAST department's environmental tracking from fragmented, manual processes into a centralized, web-based digital platform.

Process Optimization: To eliminate the institutional "validation bottleneck" by decentralizing data approval, shifting the validation authority from a single Campus Facility Manager to the respective SOAST Program Heads.

Awareness Integration: To integrate automated eco-tips and a personal carbon footprint calculator to promote student environmental responsibility.

Institutional Alignment: To generate accurate, aggregated Sustainability Reports that directly support NTC's "Alagaan ang NTC" campaign and align with the United Nations Sustainable Development Goal 13 (Climate Action).

1.2 SMART Success Criteria

To ensure the project's completion is clear, measurable, and realistic, the team will adhere to the following SMART criteria as the baseline for CarbonTrack's success:

- **Specific:** Deliver a web-based system featuring three Role-Based Access Control (RBAC) levels (SOAST Student, Program Head, School Management) focused strictly on three metrics: Travel Distance (km), Electricity Usage (hours), and Waste Volume (kg).
- **Measurable:** Achieve a 100% validation enforcement rate for reports, with system performance thresholds of ≤ 2 seconds for carbon computations, ≤ 5 seconds for report generation, and support for ≤ 100 concurrent user sessions.
- **Achievable:** Build a lightweight, web-based application relying strictly on manual numerical inputs instead of IoT, GPS, or paid third-party APIs to match the institution's desktop laboratory infrastructure and academic budget.
- **Relevant:** Provide verifiable, data-driven insights to School Management to formulate campus-wide carbon reduction policies for the "Alagaan ang NTC" initiative.

- **Time-bound:** Complete and baseline the final Software Requirements Specification (SRS) prior to the Final Exam defense in May 2026.

1.3 High-Level Risks

During the development and implementation phases of the CarbonTrack system, several high-level risks may impact its deployment or overall success. The identified risks, along with their respective mitigation strategies, are outlined below:

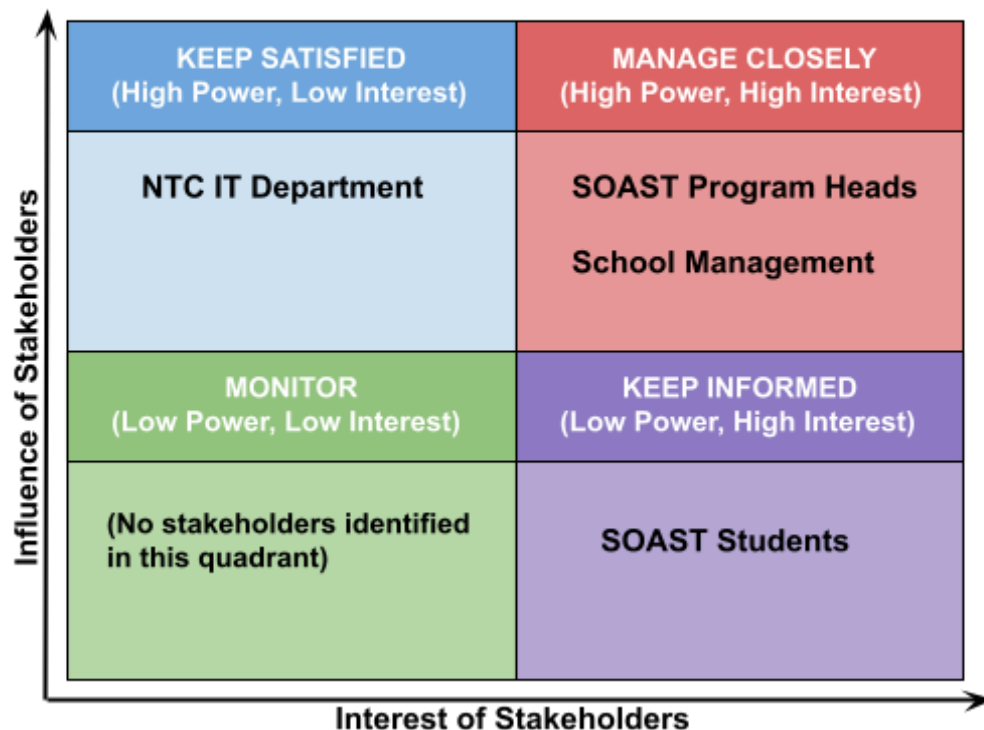
- **Data Integrity Risk (Garbage-In, Garbage-Out):** Student users may accidentally or intentionally input inaccurate, unrealistic, or fabricated environmental data (e.g., impossible travel distances or excessive waste volumes), resulting in unreliable sustainability reports.
 - *Mitigation:* Implement strict data validation rules on input fields and enforce a mandatory approval workflow via the SOAST Program Heads.
- **Infrastructure and Connectivity Risk:** System accessibility depends entirely on NTC's campus laboratory computers and network uptime. Hardware failures, power outages, or unstable internet connections may disrupt system operations.
 - *Mitigation:* Optimize the web application to be lightweight for faster loading and restrict deployment expectations to controlled campus laboratory hours.
- **Low User Adoption Risk:** Students may find regular environmental logging tedious or time-consuming, leading to resistance, low participation, and incomplete institutional data.
 - *Mitigation:* Streamline the user interface by limiting tracking to only three core metrics and automating the carbon footprint calculation to minimize manual effort.
- **Scope Creep Risk:** Stakeholders may request premature system expansion (e.g., campus-wide rollout, advanced IoT integration, or extra metrics), causing development delays and infrastructure strain.
 - *Mitigation:* Strictly enforce the project charter boundaries, limiting the initial deployment exclusively to the SOAST department

2. Stakeholder Analysis

This section identifies the key people involved in the project. It shows how much power (influence) and interest they have in the CarbonTrack system, and how the team solved their initial concerns.

2.1 Interest/Influence Matrix

The following grid visually categorizes the stakeholders based on their level of Power (Influence) and Interest in the system, which helps determine how the team should manage them.



High Influence / High Interest (Manage Closely):

- *SOAST Program Heads*: They are the ones who will approve or reject the data. If the system is hard to use, the data will not move forward.
- *School Management*: They are the end-users of the reports. They need accurate data to create rules for the "Alagaan ang NTC" campaign.

Low Influence / High Interest (Keep Informed):

- *SOAST Students*: They are the main users who will input the data. They just want the system to be fast, simple, and private.

High Influence / Low Interest (Keep Satisfied):

- *NTC IT Department:* They provide the computers and internet connection in the laboratory. The system must follow their basic security rules.

2.2 Formal Negotiation Log

During the analysis and requirements gathering stages, several operational conflicts, technical concerns, and scope-related disagreements emerged between stakeholders. The following negotiation log documents the issues raised, the decisions made by the development team, and the rationale behind each decision.

Issue	Stakeholder Concern	Final Decision	Rationale
<i>Campus-Wide Deployment Scope</i>	School Management initially proposed implementing the system across the entire NTC campus immediately.	The project scope was formally restricted to the SOAST Department only.	The development team determined that campus-wide implementation would overwhelm the validation process and increase infrastructure complexity before the system was fully stabilized. Limiting the scope allows the team to validate the workflow under manageable conditions first.
<i>Validation Authority</i>	Program Heads expressed concern that environmental records could become unreliable if students were allowed to self-approve submissions.	Only authorized SOAST Program Heads shall possess validation authority.	This decision guarantees data integrity and ensures that all environmental records included in institutional reports have passed administrative review.
<i>Manual Input Estimation Difficulties</i>	Students reported difficulty estimating travel distances and waste weight measurements accurately.	Integrated estimation references and conversion guides were added to the system.	Built-in guidance tools reduce user confusion and improve the accuracy of manually entered environmental data.

<i>System Complexity</i>	Stakeholders feared that a highly technical or complicated system would discourage student participation.	The interface was intentionally simplified to only three primary environmental inputs.	Simplicity improves usability, minimizes training requirements, and increases the likelihood of consistent student participation.
<i>Privacy Concerns</i>	Students expressed concern regarding how their environmental and personal information would be stored and used.	Mandatory Data Privacy Consent functionality was added.	Compliance with RA 10173 requires transparent disclosure regarding the collection and processing of Personally Identifiable Information (PII).
<i>Automated Tracking Requests</i>	Some stakeholders suggested GPS-based tracking and automatic data collection.	Automatic tracking features were excluded from the system scope.	GPS integration would introduce privacy risks, higher infrastructure costs, and additional technical complexity that exceeded the project's academic scope.

3. Requirements Elicitation

This section shows how the development team gathered data, interviewed project stakeholders, and identified problems in the current manual workflow.

3.1 Documented Interview Protocols/Methods

The team used two (2) main interview methods inside the National Teachers College (NTC) computer laboratories to find system rules and user problems: Individual Interviews and Focus Group Discussions (FGD).

Protocol 1: Administrative Interview Framework (Target: SOAST Program Head)

Objective: Understand the step-by-step approval process, data checking rules, required tracking records, and the final reporting needs.

Methodology: A guided one-on-one meeting with the Program Head, complete with a synchronized audio recording for accurate note-taking.

Active Question Script:
<i>“Can you walk us through the step-by-step verification process required to validate a student environmental log before it can be securely aggregated into institutional sustainability reports?”</i>
<i>“What exact system rules are necessary to prevent student users from submitting fake, exaggerated, or unverified numbers for their daily activities?”</i>
<i>“What legal standards or data privacy features must this web application actively use to protect the sensitive personal records of the department?”</i>

Protocol 2: Executive Interview Framework (Target: School Management)

Objective: Match the collected system data with the high-level school targets, sustainability campaign goals, and the official administrative reporting standards.

Methodology: A guided executive meeting with the administrative officers to understand their expectations

Active Question Script:
<i>“What specific environmental numbers must the system collect to directly support and measure the true success of the 'Alagaan ang NTC' campaign?”</i>
<i>“How should the final sustainability summaries be designed and displayed to help the management easily create new data-driven school policies?”</i>
<i>“What are the strict time limits and operating needs for generating ready-to-print sustainability</i>

trends for the official school evaluation?"

Protocol 3: Student Focus Group Framework (Target: SOAST Student Representatives)

Objective: Identify user difficulties during data entry, problems with guessing exact numbers, default browser usage, and reasons why students might quit the app.

Methodology: A face-to-face interactive workshop session attended by fifteen (15) active SOAST student representatives.

Active Question Script:

"What exact difficulties do you experience when trying to guess or calculate your actual daily travel distances and classroom waste weights by yourself?"

"What specific web browser version do you regularly use across the computer laboratory workstations whenever you access the campus web services?"

"What is the maximum amount of time you are willing to spend each day typing your environmental numbers before you feel tired and close the form?"

3.2 Raw Notes Summary

The collected answers from all three interview protocols were carefully reviewed, cleared of repeating ideas, and transformed into direct, actionable system rules.

A. Verification and Administrative Workflow Realities (Program Heads)

- **Problems with the Manual Process:** The current paper logbook format requires too much manual work. The assigned staff spend 15 to 20 minutes per student group manually matching conversion numbers against paper files before approval.
- **No Automatic Approval Rule:** Self-approval or automatic checking features for students are completely banned. All submitted entries require immediate routing to an isolated "Pending" data folder for the head's review.
- **Required Rejection Remarks:** To maintain a clear tracking history, a strict business rule is needed: rejecting a log requires mandatory justification. The action to reject should be blocked unless the Remarks box contains a typed reason.

B. Macro-Level Reporting and Strategic Alignment (School Management)

- **Total Numbers for the Campaign:** Management stated that reports must never show scattered individual logs. Instead, the requirement is to generate clear, summarized visual charts showing total kilometers, hours, and kilograms for the campaign.

- **Data Needed for School Rules:** The school leaders need historical trend graphs (like monthly changes in resource usage) to serve as a factual basis when making future campus-wide environmental rules.
- **Fast Report Generation:** Executive reporting functions require instant loading (≤ 5 seconds for summary generation) to ensure smooth data presentations during admin board meetings without waiting.

C. Student Data Capture and Client-Side Friction (Student)

- **No Tools to Measure Data:** Students show a strong desire to support the "Alagaan ang NTC" campaign but lack the tools to get exact numbers. Classrooms do not have weighing scales or tracking tools to help them.
- **Request for Help Tools:** Users asked for a built-in conversion guide directly inside the screen (like converting common trash items into kilogram numbers) and an external clickable link leading to map services.
- **Keep the Form Short:** Long and complex web forms will quickly cause users to leave the page. To ensure students log in daily, the input screen requires a strict limit of just three specific fields.

D. Infrastructure, Security, and Compliance Bounds

- **Made for Google Chrome:** The NTC computer laboratory screens are specifically set up to prioritize the desktop Google Chrome browser. All visual layouts require full optimization for this specific browser environment.
- **Slow School Internet:** The campus internet speeds change constantly during busy laboratory hours. Computations and data retrieval demand high performance (≤ 3 seconds under a concurrent load of 50 users) to avoid connection timeout errors.
- **Student Privacy Fears:** Students expressed strong fears about submitting data because of potential hacking risks. They demand a clear Data Privacy Agreement step on the login screen to ensure strict compliance with Republic Act No. 10173 (Data Privacy A

4. Specific Requirements (MoSCoW & Shall Syntax)

This section compiles the absolute baseline engineering targets for the CarbonTrack monitoring application, systematically partitioned into Functional and Non-Functional requirements using MoSCoW notation and "Shall" syntax.

4.1 Functional Requirements (FR)

ID	Priority	Raw Stakeholder Quote	Structured Requirement	Type
REQ-FR-01	Must Have	<i>"Since we use the computer lab, we should just log in using our official NTC student number."</i>	The system shall enforce account verification processes requiring an alphanumeric NTC Student ID and password sequence.	Authentication
REQ-FR-02	Must Have	<i>"Under the Alagaan ang NTC initiative, we must track only campus-related environmental impacts..."</i>	The system shall restrict all transactional data entry fields to commute distance, campus electricity usage, and campus waste volumes inside the SOAST Department.	Boundary Control
REQ-FR-03	Must Have	<i>"It's hard to estimate how many kilos of waste we have on campus because we don't have a weighing scale..."</i>	The system shall provide an in-interface Conversion Guide widget to help students accurately translate item volumes into mass weights in kilograms (kg).	User Assistance
REQ-FR-04	Must Have	<i>"The SOAST Program Head must validate if what the student submitted is okay."</i>	The system shall provide a secure Data Validation Module enabling the authenticated SOAST Program Head to approve or reject incoming pending activity items.	Data Validation
REQ-FR-05	Must Have	<i>"As School Management, we need to see a summary of all programs to know if the Alagaan ang NTC goals are met."</i>	The system shall generate an overall aggregated summary analytics report layout exclusively accessible by verified School Management accounts.	Reporting

REQ-FR-06	Should Have	<i>"I don't know exactly how many kilometers my daily trip to NTC from home is. I wish there were an easy way to check..."</i>	The system shall embed an external target shortcut hyperlink routing users outward to interactive mapping engines (e.g., Google Maps) within the input interface.	External Integration
REQ-FR-07	Could Have	<i>"It would be nice if the system gives us tips on how to lower our carbon footprint after we submit."</i>	The system shall display dynamic, context-aware eco-friendly tips on the student dashboard interface layout upon successful calculation tasks.	Enhancements
REQ-FR-08	Won't Have	<i>"What about my electricity and garbage at home? Will we include them in the system?"</i>	The system shall not track, log, or calculate any electricity patterns or refuse disposal volume attributes generated outside the NTC campus bounds.	Scope Boundary
REQ-FR-09	Won't Have	<i>"Can the system automatically detect my location so I don't have to manually type my commute distance?"</i>	The system shall not implement background GPS geo-location detection mechanisms or automated cellular real-time background tracking protocols.	Scope Boundary

4.2 Non Functional Requirements (NFR)

ID	Priority	Engineering Parameter	Structured Requirement	Metric Target
REQ-NFR-01	Must Have	Network Latency / Performance	The system shall process a Carbon Footprint score calculation and commit changes to database tables in ≤ 3 seconds under a concurrent simulated load of 50 users running commands over the standard Campus Wi-Fi connection.	≤ 3.0 Seconds Latency
REQ-NFR-02	Must Have	Data Management / Integrity	The database engine shall structurally isolate and completely exclude all logs marked as Pending	100% Data Isolation

			or Rejected from entering report processing pipelines.	
REQ-NFR-03	Must Have	Account Security / Encryption	The application logic layer shall hash all credential arrays via strict one-way hashing schemes (e.g., BCrypt) before passing updates to database memory.	Cryptographic Hashing
REQ-NFR-04	Should Have	Browser Environment Compatibility	The application client-side codebase and layout grids shall be optimized exclusively for responsive script rendering within desktop Google Chrome browser engines.	Chrome Engine Optimization
REQ-NFR-05	Should Have	Database Query Scalability	The backend relational database queries shall filter, compile, and aggregate thousands of standardized 3NF records into tabular summaries within ≤ 5.0 seconds.	≤ 5.0 Seconds Query

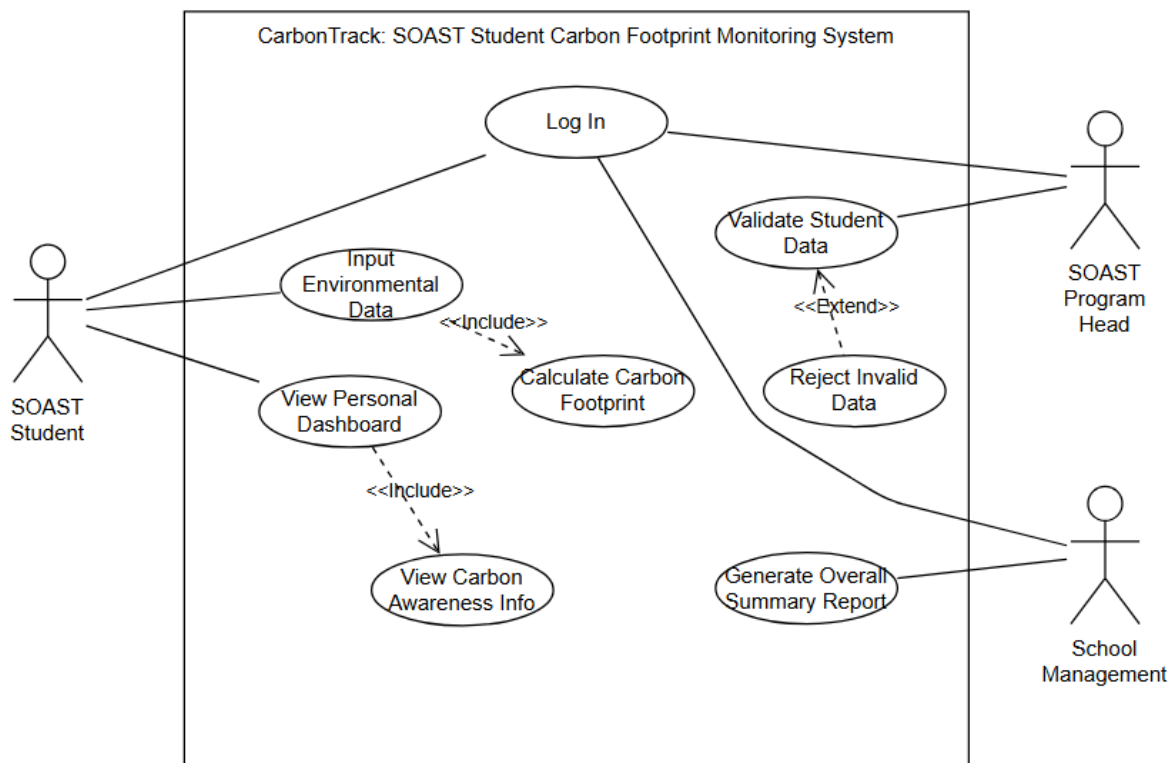
5. Logical Use Case Models

This section details the logical interaction boundaries of CarbonTrack, mapping out how system actors trigger core system transactions.

5.1 Actor Identification

Actor	Role Type	Responsibilities
SOAST Students	Primary	The main data generator who logs daily environmental metrics (commute, charging, waste).
SOAST Program Head	Secondary	The authority responsible for validating, approving, or rejecting student submissions for their specific program.
School Management	Secondary	The consumer of high-level aggregated reports used to monitor the progress of the "Alagaan ang NTC" initiative.

5.2 Use Case Diagram



Based on the CarbonTrack Use Case Diagram, the system follows these simple rules:

- **System Boundary:** The box labeled "**CarbonTrack: SOAST Student Carbon Footprint Monitoring System**" defines the scope of the application.
- **Authentication:** All actors must first perform the **Log In** use case to access their respective features.
- **Include Relationships:**
 - **Input Environmental Data** <<includes>> **Calculate Carbon Footprint:** The system automatically computes the CO2 equivalent immediately upon data submission.
 - **View Personal Dashboard** <<includes>> **View Carbon Awareness Info:** Eco-friendly tips and information are automatically integrated into the student's dashboard view.
- **Extend Relationship:**
 - **Reject Invalid Data** <<extends>> **Validate Student Data:** Rejection is a conditional step that only occurs if the Program Head identifies erroneous or illogical entries.

5.3 Use Case Narrative

Field	Detail
Use Case Name	Input Environmental Data
Primary Actor	SOAST Student
Pre-conditions	1. The student is logged into the system via Google Chrome in the SOAST computer lab. 2. The student has not yet submitted data for the current date.
Success Scenario	1. The student accesses the data entry form. 2. The student provides the following inputs: ● Travel Distance: The student enters distance in km (User may click the provided Google Maps reference link to verify the exact distance). ● Electricity Usage: The student selects or enters the number of hours spent charging devices on campus. ● Waste Volume: The student enters the weight in kg (User may refer to the on-screen Conversion Guide to estimate weight based on item counts like bottles or wrappers). 3. The student clicks "Submit."

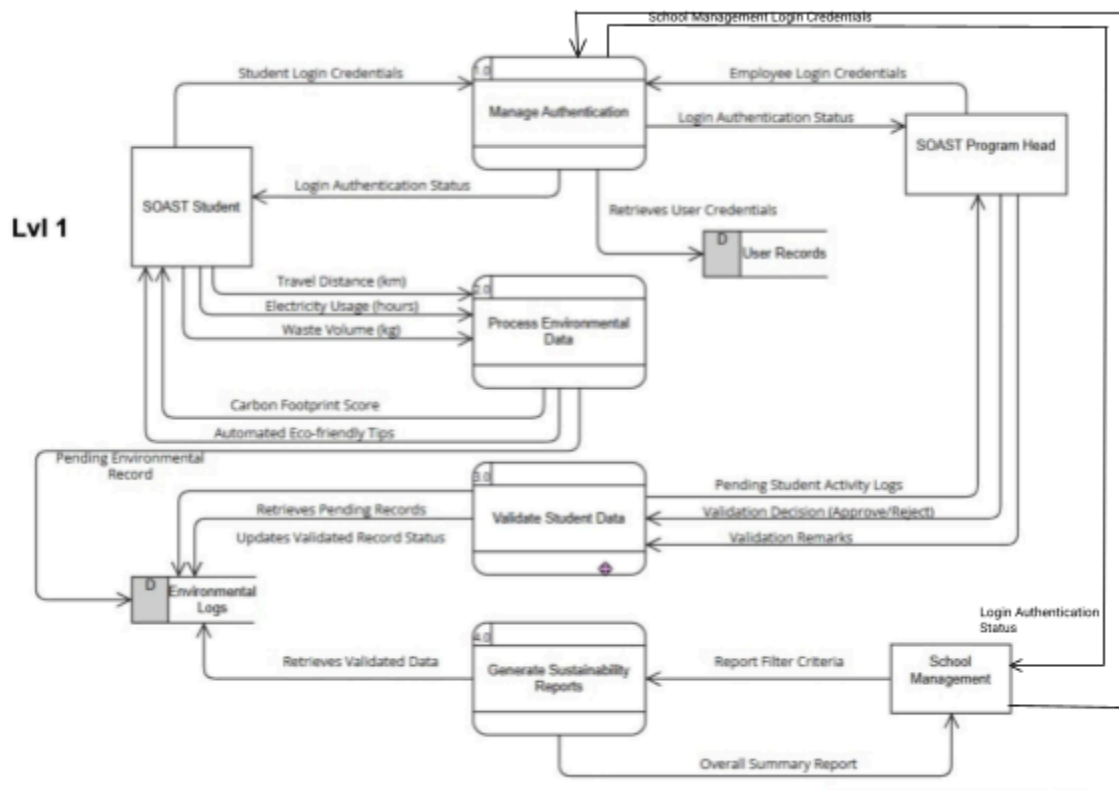
	4. (Include) The system calculates the carbon footprint score. 5. The system saves the record as "Pending" and updates the dashboard. 6. The system displays a success message with Carbon Awareness tips
Post-conditions	The environmental record is persisted in the database and flagged for Program Head review.

6. System Boundary

This section establishes the absolute data limits and operational perimeter of CarbonTrack, mapping out how external entities exchange information with the core processing layer.

6.1 Level 1 DFD Decomposition

The systemic execution behavior of the CarbonTrack monitoring core is partitioned into four major functional process blocks, working over two centralized, persistent data stores.



Process Architecture Elements

System Model: Level-1 Data Flow Diagram (DFD).

External Entities: SOAST Student, SOAST Program Head, School Management.

Data Stores: D1: User Records, D2: Environmental Logs.

Processes:

Process 1.0: Manage Authentication.

Process 2.0: Process Environmental Data.

Process 3.0: Validate Student Data.

Process 4.0: Generate Sustainability Reports.

Detailed Process Specifications and Structural Data Flows

1.0 Manage Authentication

- Student Login Flow: Process 1.0 accepts Student Login Credentials from the SOAST Student interface.
- Employee Login Flow: Process 1.0 accepts Employee Login Credentials from the SOAST Program Head interface.
- School Management Login Flow: Process 1.0 accepts School Management Login Credentials from the School Management interface.
- Database Lookup: Process 1.0 executes a retrieval query to fetch matching security keys from data store D1: User Records.
- Response Dispatch: Process 1.0 returns the verified Login Authentication Status back to the SOAST Student, SOAST Program Head, and School Management interfaces to control dashboard routing.

2.0 Process Environmental Data

- Metric Capture Flow: Process 2.0 captures three data streams from the SOAST Student entry form: Travel Distance (km), Device Usage (hours), and Waste Volume (kg).
- Database Record Lock: Process 2.0 immediately writes a transactional entry labeled as a Pending Environmental Record to data store D2: Environmental Logs.
- Response Generation: Process 2.0 runs the background algorithms to instantly return the calculated Carbon Footprint Score and context-specific Automated Eco-friendly Tips to the SOAST Student dashboard view.

3.0 Validate Student Data

- Queue Retrieval: Process 3.0 continuously queries data store D2: Environmental Logs to fetch log rows flagged with an unverified status.
- Dashboard Population: Process 3.0 populates the administrative interface by transmitting the Pending Student Activity Logs to the SOAST Program Head.
- Decision Capture: Process 3.0 intercepts the supervisor's input variables consisting of the Validation Decision (Approve/Reject) and the mandatory text strings inside the Validation Remarks field.
- Database Commits: Process 3.0 executes an update query to modify the Validated Record Status inside data store D2: Environmental Logs, locking the final status parameters.

4.0 Generate Sustainability Reports

- Query Filtering: Process 4.0 captures strict execution targets defined as Report Filter Criteria from the School Management reporting module.
- Data Aggregation: Process 4.0 triggers an optimized lookup script to retrieve Validated Data from data store D2: Environmental Logs.
- Reporting Output: Process 4.0 compiles the parsed mathematical database metrics into a high-level visual and tabular layout, sending the finished Overall Summary Report back to School Management.